SIMATS ENGINEERING
Department of Computer Science and Engineering
CSA16 – Data Warehousing and Data Mining
ASSIGNMENT 2 – CO2

Design, Build and Validate a Data Warehouse and OLAP Reporting Solution for a Retail Chain

CO2: Apply suitable Data Pre-processing techniques like Data cleaning, Data integration, Data transformation and data Reduction for effective data preparation

Student Name	Register Number	Department / Section	Academic Year / Semester	Faculty Name	Nature of Assignment	Duration	Total Marks	Date of Submission	Date of Evaluation
Harini S	192511153	CSE	2026–2027	Dr.D. Beulah David	Individual	1 Week	100		

MARK SHEET (Rubric-based Evaluation)

Task	Task Title	Max	Marks Awarded	Remarks / Evidence Expected
Q1	Mining Functionalities and Algorithm Mapping	10		Six telecom problems mapped to functionality, supervised/unsupervised type, data attributes/source and at least two suitable algorithms; evaluate the all-purpose algorithm proposal.
Q2	KDD Process, Architecture and Mining Issues	10		MediCore KDD mapping; labelled mining-system architecture; classification by four views; at least five issue–mitigation pairs; distinguish mining from SQL and basic statistics.
Q3	Data Dispersion and Outlier Analysis	14		Mean, median, mode, midrange, range, quartiles, IQR, variance and standard deviation; five-number summary, boxplot, outlier test, removal analysis, measure classification and tool verification.
Q4	Data Cleaning and Noise Handling	12		Defect classification and justified handling; equal-frequency binning (depth 4), mean/boundary smoothing; regression/clustering discussion and imputation impact.
Q5	Data Integration and Transformation	14		Conflict-resolution schema; redundancy; Pearson correlation and chi-square calculations; min-max, z-score and decimal scaling; Euclidean distance comparison and transformation examples.
Q6	Data Reduction and Discretization	14		Reduction methods, sampling comparison for rare failures, PCA criterion, equal-width/equal-frequency intervals, entropy/information-gain table and discretization comparison.
Q7	Day-4 Design Challenge – Diagnose, Redesign and Validate	26		A: diagnose pipeline defects (2); B: measurable objectives/constraints (2); C: revised preprocessing pipeline and justification (4); D: predictions (1); E: implementation/validation (2); F: defence and remaining risk (2).
	TOTAL	**100**		**Marks to be awarded by the faculty.**

OVERALL REMARKS

Faculty feedback:

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Student Signature	Faculty Signature	Program Director
		

MANDATORY SUBMISSION EVIDENCE CHECK

✓ = present P = partly present ✗ = missing. Mark the status after checking the student's submitted report and supporting evidence.

Task	Evidence Expected	Status	Note
Q1	Six problems mapped to functionality, learning type, data attributes/source and candidate algorithms.	☐	All six cases and the design decision addressed.
Q2	KDD activities, architecture diagram, four-view classification and issue–mitigation table.	☐	At least five scenario-specific issues; SQL/statistics comparison.
Q3	Manual calculations, five-number summary, boxplot, outliers and tool verification.	☐	Quartile/variance conventions stated; removal analysis included.
Q4	Defect classification, handling strategy, binning and smoothing calculations.	☐	All bins shown; imputation impact discussed.
Q5	Conflict table, correlation, chi-square expected values and interpretation.	☐	Three normalisation methods and distance comparison included.
Q6	Reduction strategy, sampling choice, PCA criterion and discretisation calculations.	☐	Intervals and information-gain calculations included.
Q7-A/B	Pipeline diagnosis, violated principles, measurable objectives and constraints.	☐	All original pipeline defects considered.
Q7-C/D	Ordered revised pipeline, alternatives, training-only decisions and predictions.	☐	Justifications and predictions documented.
Q7-E/F	Implementation comparison, metrics, interpretation, defence and remaining risk.	☐	Same test data used; default recall reported.

IMPLEMENTATION EVIDENCE CHAIN (Tasks 7–9)

Evidence Item	Status	Observation
Manual calculation sheets	☐	Formulae, assumptions and intermediate values shown for Q3–Q6.
Dataset / data source	☐	Dataset description, source or link and relevant attributes documented.
Spreadsheet / Python / Weka / KNIME files	☐	Files or workflows used for calculation and verification submitted.
Tool settings and screenshots	☐	Settings, selected attributes, preprocessing and outputs captured.
Plots and visual evidence	☐	Boxplot, architecture and other required diagrams are labelled and readable.
Q7 baseline pipeline	☐	Original preprocessing steps and baseline results are reproducible.
Q7 redesigned pipeline	☐	Ordered steps, fitted parameters and training/test separation documented.
Validation and metric comparison	☐	Same test set; accuracy and class-sensitive measures, especially default recall, reported.
Reproducibility and consistency	☐	Dataset, code, calculations, screenshots and report values agree.

SUBMISSION CHECKLIST (not separately marked)

Requirement	Status
All Questions Q1–Q7 are answered.	☐ Yes ☐ Partly ☐ No
All six Q1 problems are mapped to mining functionalities and algorithms.	☐ Yes ☐ Partly ☐ No
KDD stages, system architecture and mining issues are explained using the hospital scenario.	☐ Yes ☐ Partly ☐ No
Q3 manual calculations, quartile method, boxplot and outlier analysis are included.	☐ Yes ☐ Partly ☐ No
Q4 defects, cleaning decisions, binning and smoothing calculations are shown.	☐ Yes ☐ Partly ☐ No
Q5 integration conflicts, correlation, chi-square and normalisation calculations are included.	☐ Yes ☐ Partly ☐ No
Q6 reduction methods, sampling, discretisation and information-gain calculations are included.	☐ Yes ☐ Partly ☐ No

Q7 diagnosis, objectives, revised pipeline, predictions, implementation and defence are provided.	☐ Yes ☐ Partly ☐ No
Datasets, spreadsheets, code/workflows and tool screenshots are submitted where applicable.	☐ Yes ☐ Partly ☐ No
Diagrams, tables, calculations and references are numbered, labelled and readable.	☐ Yes ☐ Partly ☐ No
Assumptions, formulas, conventions and sources are clearly stated.	☐ Yes ☐ Partly ☐ No
No unfilled template placeholders remain in the student report.	☐ Yes ☐ Partly ☐ No
Viva recommended where calculations, implementation or results need clarification.	☐ Yes ☐ No

Note: This DWDM assignment evaluation template is based on Assignment 2 – CO2. Blank marks and evidence statuses are intentionally left for faculty evaluation.

IMPORTANT CALCULATION CONVENTIONS

- Q3 quartiles: use the median-of-halves method. For an even number of observations, divide the ordered data into two equal halves and find each half's median.
- Q3 variance: report population variance and standard deviation for the supplied 24 observations; sample estimates are also shown for reference.
- Q4 equal-frequency bins: depth 4 means four bins, with three values in each bin because 12 values are supplied.
- Q5 z-score normalisation: use population standard deviation. Decimal scaling uses j = 6 because the largest absolute income is 150,000 and 150,000 / 10^6 < 1.
- Q6 entropy: binary class entropy uses base-2 logarithms. Candidate split points are midpoints between adjacent values as instructed.

Where the assignment asks for tool verification or implementation results, the report gives the correct procedure and clearly marked result templates. Actual Weka/KNIME/Python screenshots and measured model metrics must be generated by running the student's implementation; they are not invented here.

Q1 – MATCHING BUSINESS PROBLEMS TO MINING FUNCTIONALITIES (5 MARKS)

1.1 Mapping the six business problems

Problem	Mining functionality	Learning type / output	Required attributes and source	Candidate algorithms and reason
P1: Plans and add-ons bought together	Association analysis	Unsupervised; frequent itemsets and rules such as {Plan A} → {Roaming Pack}.	Subscriber/order basket ID, plan ID, add-on IDs, date, channel. Warehouse subscription/billing transactions.	Apriori: clear support-confidence rule generation; FP-Growth: avoids repeated candidate generation and is efficient for large baskets.
P2: Predict customer churn next month	Classification / prediction	Supervised; probability or class label: churn / no churn.	Historical churn label, tenure, recharge amount, complaints, dropped calls, data usage trend, plan, payment delays. Warehouse with time-consistent snapshots.	Decision Tree: interpretable churn rules; Random Forest: combines trees and captures non-linear interactions. Logistic Regression is also a useful baseline.
P3: Natural subscriber usage segments	Clustering	Unsupervised; groups such as low-data, voice-heavy, high-value users.	Monthly data, voice minutes, SMS, recharge, tenure, plan, location. Warehouse customer-month feature table.	K-Means: efficient for large numeric datasets after scaling; DBSCAN: discovers irregular clusters and can mark noise, but needs parameter tuning.
P4: SIM cards with unusual calling behaviour	Outlier analysis	Usually unsupervised; anomaly score or flagged SIMs.	Call frequency, duration, destinations, time-of-day, roaming, geographic changes, failed calls. Call Detail Records (CDR) warehouse.	Isolation Forest: scalable anomaly scoring; Local Outlier Factor (LOF): compares local density and detects locally unusual behaviour.
P5: Characteristics of high-value vs low-value customers	Concept/class description (characterisation and discrimination)	Descriptive; profiles common traits and contrasts two predefined groups.	Customer lifetime value, ARPU, tenure, plan, usage, recharge, complaints, add-ons. Customer and billing warehouse.	Attribute-oriented induction / OLAP roll-up: summarises profiles by concept levels; Decision Tree: produces readable rules that distinguish high- and low-value groups.
P6: Usage change over 24 months	Evolution / trend analysis	Time-oriented; trends, seasonality, growth/decline and change points.	Monthly data usage per customer, month, plan changes, promotions, season, region. Time-series warehouse.	Moving averages / exponential smoothing: reveal smooth trend and seasonality; ARIMA: models autocorrelation and forecasts future usage.

1.2 Why one all-purpose algorithm is unsuitable

The manager's proposal should be rejected. The six problems ask for fundamentally different outputs: association rules, labelled predictions, unlabelled groups, anomaly scores, descriptive profiles and temporal trends. Apriori requires transaction baskets; a classifier requires labelled examples; clustering needs a similarity measure; outlier detection focuses on rare deviations; and evolution analysis uses ordered time observations. No single algorithm optimises all these objectives.

- Using classification for P3 would require labels that do not exist, contradicting the requirement to discover natural segments.
- Using clustering for P2 would not directly estimate the probability that a known customer will churn next month.
- Using association rules for P4 may find co-occurrences but will not reliably rank abnormal SIM behaviour.

TeleNova should build an algorithm portfolio selected by business objective, data type, scale, interpretability and validation metric. Common preprocessing and feature stores can be shared, but the mining algorithms and evaluation methods must remain task-specific.

Q2 – THE KDD PROCESS, SYSTEM ARCHITECTURE AND MINING ISSUES (5 MARKS)

2.1 KDD steps for MediCore

KDD step	MediCore activity	Example / output
1. Data selection	Select eligible hospital admissions and link records from admission, laboratory, pharmacy and billing systems. Define readmission as a new admission within 30 days after discharge, excluding planned returns if policy requires.	Patient/admission cohort with unique patient and encounter identifiers.
2. Preprocessing	Resolve duplicate patient IDs, missing values, inconsistent units, coding differences, impossible dates and conflicting records. Protect identifiers.	Clean, integrated and quality-checked patient table.
3. Transformation	Create features such as age group, previous admissions, length of stay, lab summaries, medication count, diagnosis groups and discharge destination. Create target label using future 30-day outcome.	One row per index admission, with predictors measured before discharge and a 30-day readmission label.
4. Data mining	Train and compare classification models; optionally use association analysis to find combinations of clinical and service factors.	Readmission risk score, important predictors and candidate patterns.
5. Interpretation / evaluation	Evaluate on a time-separated test set; examine recall, precision, PR-AUC, calibration and subgroup performance. Clinicians review whether patterns are clinically meaningful.	Validated findings, limitations and decision-support report.

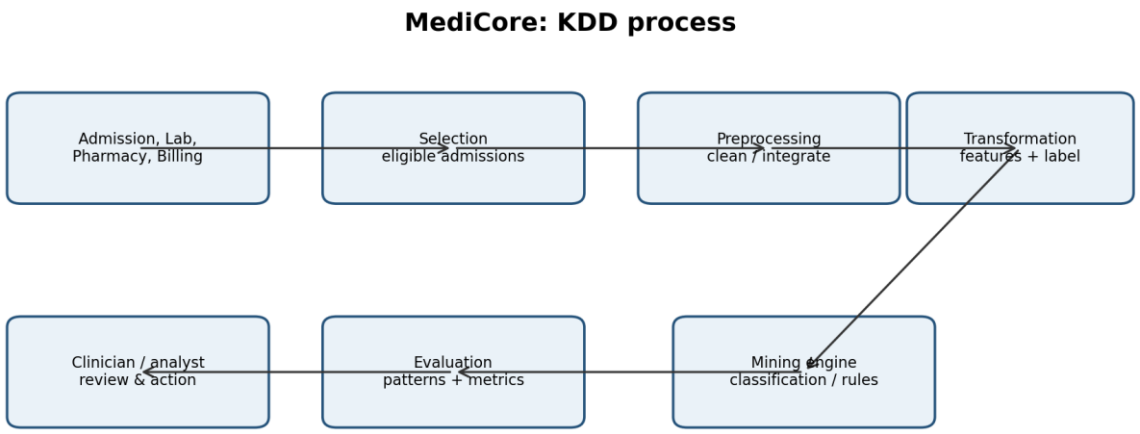


Figure 1: KDD process mapped to MediCore's hospital data.

2.2 Typical data mining system architecture

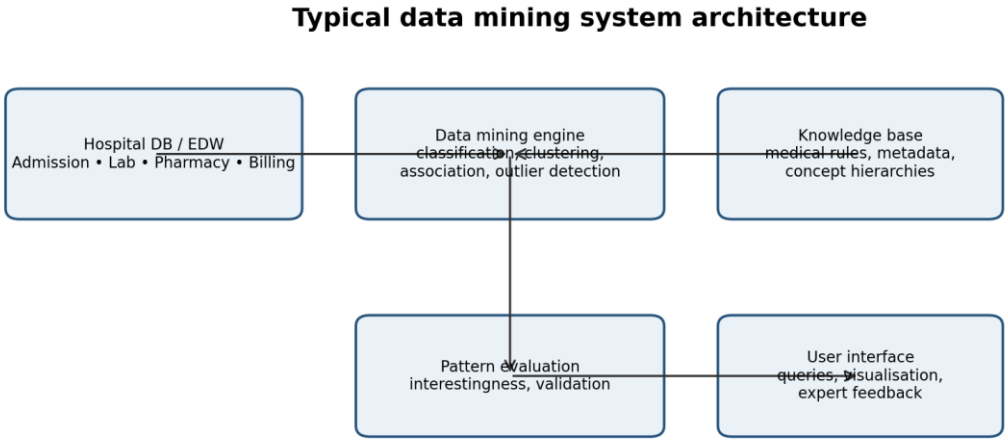


Figure 2: Data mining system architecture and the role of each component.

Component explanation:

- Database / data warehouse server: stores integrated admission, laboratory, pharmacy and billing data. It provides cleaned, governed data to the mining engine.
- Data mining engine: contains algorithms for classification, clustering, association, prediction and outlier analysis.
- Knowledge base: stores medical terminology, diagnosis hierarchies, metadata, constraints and previously accepted patterns. It guides mining and interpretation.
- Pattern evaluation module: ranks patterns using measures such as support, confidence, lift, accuracy, recall and statistical significance. It filters uninteresting patterns.
- User interface: allows analysts and authorised clinicians to select data, configure mining tasks, inspect visualisations and provide expert feedback.

2.3 Classification of the MediCore mining system

View	Classification for MediCore	Reason
Kinds of data	Relational, transactional, time-series and potentially text data.	Admission, laboratory, pharmacy and billing tables contain structured records; clinical notes may contain text.
Kinds of knowledge	Classification/prediction, association, characterisation, clustering and outlier patterns.	Main target is 30-day readmission prediction, supported by explanatory patterns and patient subgroups.
Techniques used	Machine learning, statistical learning, database/OLAP methods and visual analytics.	Models identify complex relationships; SQL/OLAP integrates and aggregates; visual tools help clinicians inspect results.
Applications	Healthcare analytics, readmission risk management, resource planning and quality improvement.	The output can help authorised care teams identify patients who may need additional discharge support.

2.4 Data mining issues and mitigations

Issue category	MediCore-specific risk	Mitigation
Methodology and user interaction	A mathematically strong model may identify correlations that clinicians cannot interpret or act upon.	Involve clinicians in defining the target and features; provide explanations, confidence and clinically reviewed reports.
Data quality / integration	Different patient identifiers, units, diagnosis codes and timestamps may produce incorrect joins.	Use a master patient index, standard code mappings, unit conversion, validation rules and data lineage.
Efficiency and scalability	Millions of laboratory and billing events can make repeated mining slow.	Use a warehouse, indexed keys, partitioning, incremental refresh, feature aggregation and scalable algorithms.
Diversity of data types	Structured lab values, medication events, diagnosis codes and free-text notes have different formats.	Use type-specific preprocessing, text extraction only where approved, and a documented common feature model.
Privacy and social issues	Health data is highly sensitive; risk scores may create unfair treatment or stigma.	Apply role-based access, encryption, de-identification where possible, audit logs, consent/legal review and fairness checks.
Temporal leakage	A feature recorded after discharge may reveal the future readmission outcome.	Use only information available at the prediction time; use time-based validation and strict feature timestamps.
Class imbalance	Readmissions may be less frequent than non-readmissions, so accuracy may hide poor detection.	Use recall, precision, PR-AUC, class weights and threshold selection; validate on realistic prevalence.

2.5 SQL, statistics and data mining are different

SQL retrieves data that a user already knows how to ask for. For example, SELECT COUNT(*) FROM Admissions WHERE readmitted_30_days='Yes' gives the number of readmissions. A statistical summary may calculate the average length of stay for readmitted and non-readmitted groups. These are useful but do not automatically discover complex combinations of predictors.

A mining task can train a decision tree on previous admissions and discover a pattern such as: patients with multiple previous admissions, a long stay and certain discharge conditions have a higher estimated risk of readmission. The model can then score a new eligible admission. Thus, SQL retrieves and aggregates known fields, statistics summarise selected relationships, while data mining searches many attributes for predictive or previously unknown patterns. The model identifies associations, not necessarily causal factors; clinical review is required.

Q3 – DESCRIBING AND COMPARING DATA DISPERSION (7 MARKS)

3.1 Central tendency

Ordered data (n = 24): 28, 31, 33, 35, 35, 36, 38, 40, 41, 42, 42, 43, 45, 46, 48, 50, 52, 53, 55, 58, 61, 64, 72, 95.

Mean = $\Sigma x / n = 1143 / 24 = 47.625$ minutes.

Median = average of 12th and 13th values = $(43 + 45) / 2 = 44$ minutes.

Mode = 35 and 42 minutes; both occur twice, so the data is bimodal.

Midrange = (minimum + maximum) / 2 = (28 + 95) / 2 = 61.5 minutes.

The mean (47.625) is greater than the median (44), indicating a right-skewed distribution. The large delivery time of 95 minutes pulls the mean upward.

3.2 Dispersion calculations

Range = 95 − 28 = 67 minutes.

Using the median-of-halves convention: lower half has 12 values, so Q1 = (6th + 7th)/2 = (36 + 38)/2 = 37. Upper half has 12 values, so Q3 = (6th + 7th of upper half)/2 = (53 + 55)/2 = 54.

IQR = Q3 − Q1 = 54 − 37 = 17 minutes.

Population variance: $\sigma^2 = \Sigma(x - \text{mean})^2 / n = 5{,}119.625 / 24 = 213.318$ min².

Population standard deviation: $\sigma = \sqrt{213.318} = 14.605$ minutes. If the 24 observations are treated as a sample, $s^2 = 222.592$ and $s = 14.920$ minutes (divide by n−1).

3.3 Five-number summary and outliers

Minimum	Q1	Median	Q3	Maximum
28	37	44	54	95

Lower fence = Q1 − 1.5×IQR = 37 − 25.5 = 11.5. Upper fence = Q3 + 1.5×IQR = 54 + 25.5 = 79.5. Any observation below 11.5 or above 79.5 is an outlier. Therefore, 95 minutes is an upper outlier.

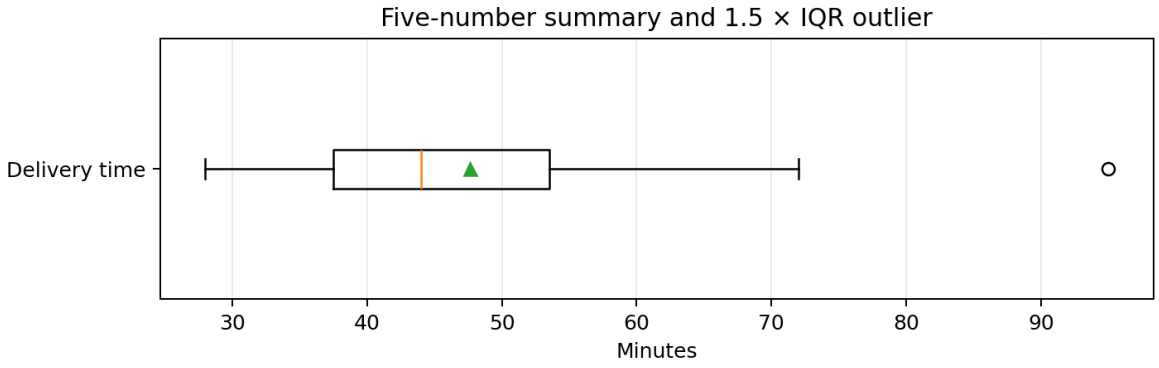


Figure 3: Boxplot of delivery times. The point at 95 minutes lies beyond the upper 1.5×IQR fence.

3.4 Recalculation after removing 95

New n = 23 and total = 1,048. Mean = 1,048 / 23 = 45.565 minutes. Median = 12th value = 43 minutes. Mode remains 35 and 42. Midrange = (28 + 72)/2 = 50 minutes. Range = 72 − 28 = 44 minutes.

Using median-of-halves for odd n and excluding the overall median: Q1 = 36 and Q3 = 53; IQR = 17. Population variance = 120.767 min² and population standard deviation = 10.989 minutes.

Measure	All 24 values	After removing 95	Change / observation
Mean	47.625	45.565	Falls by 2.060 minutes.
Median	44	43	Falls by 1 minute.
Mode	35 and 42	35 and 42	Unchanged.
Midrange	61.5	50	Falls by 11.5 minutes.
Range	67	44	Falls by 23 minutes.
Q1 / Q3	37 / 54	36 / 53	Each falls by 1 minute.
IQR	17	17	Unchanged.
Population variance	213.318	120.767	Falls by about 43.4%.
Population SD	14.605	10.989	Falls by about 24.8%.

The range and midrange change substantially because both depend directly on the maximum. Variance and standard deviation also fall because 95 is far from the centre. The median, quartiles and mode are more resistant to an extreme observation.

3.5 Classification of measures for distributed data

Measure	Class	Reason
Mean	Algebraic	Can be computed from partial count and partial sum, then combined.
Median	Holistic	Cannot be exactly combined from only local medians; generally needs ordered data or richer summaries.
Variance	Algebraic	Can be combined using count, sum and sum of squares (or equivalent sufficient statistics).
Standard deviation	Algebraic (derived)	Calculate from globally combined variance; local SDs alone cannot simply be averaged.
Range	Distributive	Global minimum is minimum of partition minima; global maximum is maximum of partition maxima, so range can be derived.

Distributive measures can be combined from the same measure on partitions (for example MIN, MAX, COUNT, SUM). Algebraic measures need a fixed-size set of summary values (for example count, sum and sum of squares). Holistic measures may require all values or an unbounded summary. This matters when warehouse data is split across many partitions.

3.6 Tool verification and typical delivery promise

Verification procedure: enter the 24 values into Excel or Python; use AVERAGE, MEDIAN, MODE.MULT, MIN, MAX, QUARTILE and VAR.P/STDEV.P; compare the output with the manual calculations. Python can use numpy.mean, numpy.median, numpy.var(ddof=0) and numpy.std(ddof=0). Quartile values may differ if a tool uses an interpolated percentile convention rather than median-of-halves. State the convention instead of treating the difference as an error.

Recommendation: publish the median of 44 minutes as the typical delivery time because it is resistant to the 95-minute outlier and better represents the central customer experience. Also report the 90th percentile or an agreed high-percentile service time, plus the IQR or a range, because customers and operations need to understand delivery variability. A promise should be based on a service-level percentile and validated over a larger, representative period, not on this small sample alone.

Q4 – CLEANING NOISY AND INCOMPLETE DATA (6 MARKS)

4.1 Defects in the sample extract

Meter	Observed issue	Classification	Recommended handling
M102	kWh is missing.	Missing value	Impute using median/mean for comparable household and time period if justified; otherwise use model-based estimate and add a missingness flag.
M103	Region is 'north' rather than 'North'.	Inconsistency	Trim and standardise region labels using a controlled lookup table.
M104	kWh = −14.	Invalid value / possible sensor fault	Do not treat as a real negative consumption reading unless export/import semantics allow it. Flag and investigate; set missing only after validation, then impute cautiously.
M105	Household type is missing.	Missing value	Look up meter registry; if unavailable, use 'Unknown' rather than guessing a class.
M105	Voltage = 2300 V while other readings are around 230 V.	Likely noise / invalid value / unit or decimal error	Apply physical range validation; quarantine and check source. Do

			not automatically convert to 230 without evidence.
M106	Region is 'S.' rather than 'South'.	Inconsistency / abbreviated value	Map approved abbreviation S. to South using reference data.

M101 (5 kWh, 230 V, North) is plausible. A reading of 5 kWh is low but not automatically invalid; consumption varies by household and hour. In contrast, 2300 V is far outside the expected domestic voltage range and may indicate a scale, decimal, or sensor problem.

4.2 Equal-frequency binning (depth 4)

Bin	Values	Bin mean	Smoothed by boundaries
Bin 1	[5, 9, 11]	8.33	[5, 11, 11]
Bin 2	[14, 18, 19]	17.00	[14, 19, 19]
Bin 3	[23, 26, 27]	25.33	[23, 27, 27]
Bin 4	[30, 33, 38]	33.67	[30, 30, 38]

Bin means smoothing replaces each value by its bin mean: 8.33, 8.33, 8.33; 17, 17, 17; 25.33, 25.33, 25.33; 33.67, 33.67, 33.67.

Boundary smoothing replaces each value by the nearest boundary (minimum or maximum of its bin). For the first bin [5, 9, 11], 9 is equally distant from 5 and 11; either boundary may be selected if a consistent tie rule is stated. The table uses the lower boundary for ties.

4.3 Regression and clustering for noise

- Regression: predict a sensor value from related variables such as previous hour's reading, temperature, household type and time of day. A large residual between actual and predicted value flags a possible noisy reading. The prediction should be used only after validation.
- Clustering: group normal meter readings by similar usage profiles. A point far from its cluster centre or in a low-density region can be flagged as unusual. Clustering is a detection aid, not proof that a reading is wrong.

The −14 kWh value may be an invalid measurement or could represent a special net-export convention; check the meter's data definition. The 2300 V value is an extreme physical-range violation and should be quarantined for source investigation, not smoothed like an ordinary small variation.

4.4 Why global-mean imputation can be harmful

Replacing every missing value with the global mean reduces observed variance because each imputed value is placed at the centre of the distribution. It can weaken correlations, hide group differences, create an artificial peak at the mean and mislead clustering or classification. It should be avoided when missingness differs by household type, region or time, when values are strongly skewed, or when missingness itself is informative. Prefer class-wise or model-based imputation with a missingness indicator, fitted only on training data in predictive modelling.

Q5 – INTEGRATION AND TRANSFORMATION OF DATA (7 MARKS)

5.1 Part A – Integration conflicts and resolution

Conflict	Admissions source	LMS source	Integrated solution
Entity identification	student_id	sid	Create a canonical StudentKey and a cross-reference table mapping each source identifier to it. Validate duplicates before merging.
Naming conflict	name	full_name	Map both to canonical attribute StudentName; standardise whitespace and case without losing original source values.
Format / unit conflict	dob may be DD-MM-YYYY	LMS may use another date format	Parse using source-specific formats and store DOB as a standard DATE

			type (ISO internally).
Value conflict	hostel_status may be Y/N	No direct equivalent; LMS may infer campus mode	Use controlled values: Hostel / Day Scholar / Unknown. Do not infer hostel status from attendance.
Scholarship conflict	scholarship	scholarship_flag	Map to ScholarshipStatus with values Yes / No / Unknown; reconcile conflicts using authoritative Admissions record and retain source lineage.
Duplicate attributes	student name and scholarship indicators may repeat	full_name and scholarship_flag overlap	Retain one governed canonical field plus source lineage; avoid storing competing copies without a reason.

Integrated schema example: Student(StudentKey, student_id, sid, StudentName, DOB, HostelStatus, ScholarshipStatus, AttendancePct, Marks, SourceSystem, LastUpdated). A separate source-key mapping table prevents accidental merging of two different students.

5.2 Redundancy and correlation

Integration can create redundancy when two source systems describe the same real-world entity or attribute using different names. For example, name and full_name may both represent a student's name. Redundant attributes increase storage, create update anomalies and can give a mining model duplicate signals. Correlation analysis helps identify strongly related numeric attributes; domain knowledge and functional-dependency checks are still required because correlation alone does not prove that two columns are duplicates.

5.3 Pearson correlation coefficient

Attendance = {60, 70, 75, 80, 85, 90}; Marks = {52, 58, 66, 70, 79, 85}. Pearson $r = \Sigma[(x-\bar{x})(y-\bar{y})] / \sqrt{[\Sigma(x-\bar{x})^2 \Sigma(y-\bar{y})^2]}$.

Using the six paired observations gives $r = 0.9851$ (approximately). This is a very strong positive linear correlation: higher attendance is associated with higher marks in this small sample. It does not prove that attendance alone causes higher marks.

5.4 Chi-square test: hostel status and scholarship

Observed table: Hostel scholarship=120, no scholarship=180, total=300; Day scholar scholarship=80, no scholarship=320, total=400. Column totals are 200 and 500; grand total = 700.

Expected frequency E = (row total × column total) / grand total.

Cell	Observed O	Expected E
Hostel / Scholarship	120	300×200/700 = 85.714
Hostel / No scholarship	180	300×500/700 = 214.286
Day scholar / Scholarship	80	400×200/700 = 114.286
Day scholar / No scholarship	320	400×500/700 = 285.714

$\chi^2 = \Sigma (O-E)^2/E = 33.60$. Degrees of freedom = (2−1)(2−1) = 1. Since 33.60 > critical value 3.84 at 0.05 significance, reject the null hypothesis of independence. There is a statistically significant association between hostel status and scholarship in the observed data. This does not establish causation.

5.5 Part C – Normalisation of annual family income

Min = ₹20,000; Max = ₹150,000; Mean = ₹67,000; population SD = ₹46,000.

Income	Min-max (0–1)	Z-score	Decimal scaling (j=6)
₹20,000	0.000000	-1.021739	0.020
₹35,000	0.115385	-0.695652	0.035
₹50,000	0.230769	-0.369565	0.050
₹80,000	0.461538	0.282609	0.080
₹150,000	1.000000	1.804348	0.150

Min-max formula: x' = (x−20,000)/(150,000−20,000). Z-score formula: z = (x−67,000)/46,000. Decimal scaling: x' = x/10^6. Decimal scaling uses j=6 because the largest absolute value has six digits and the scaled maximum is 0.15.

5.6 Euclidean distance before and after normalisation

Students A=(19 years, ₹35,000) and B=(21 years, ₹80,000).

Before normalisation: d = √[(21−19)² + (80,000−35,000)²] = √[4 + 2,025,000,000] ≈ 45,000.000 minutes/rupees in mixed units. The income scale dominates the age difference.

Min-max ranges: age 17–25; income ₹20,000–₹150,000. A'=(0.25, 0.115385); B'=(0.50, 0.461538).

After normalisation: d' = √[(0.50−0.25)² + (0.461538−0.115385)²] ≈ 0.4270.

This demonstrates that distance-based algorithms such as K-Means and KNN are sensitive to measurement scales. Normalisation prevents income's large numeric units from overwhelming age, although the selected feature weights should still reflect the business problem.

5.7 Aggregation, generalisation and attribute construction

- Aggregation: calculate each student's average monthly attendance or total annual fee payment from lower-level records.
- Generalisation: replace exact DOB with age group (18–20, 21–23, etc.) or individual hostel building with campus region.
- Attribute construction: derive AcademicPerformanceBand from marks, or AttendanceRisk = 100 − AttendancePct. The derivation rule must be documented.

Q6 – DATA REDUCTION AND DISCRETIZATION (7 MARKS)

6.1 Part A – Reduction strategies

Category	Technique	Use for 200 attributes and 5M rows/month
Data-cube aggregation	Aggregate hourly sensor values to minute/hour/day summaries, while retaining raw readings for failure investigations.	Suitable for dashboards and long-term trend mining; avoid aggregation that hides short failure spikes.
Attribute subset selection	Forward selection, backward elimination or decision-tree feature importance.	Remove irrelevant or redundant sensor channels; validate selected features against failure recall.
Dimensionality reduction	Principal Component Analysis (PCA).	Transform correlated sensor measurements into fewer components; standardise features first and inspect whether rare-failure signals are retained.
Numerosity reduction	Stratified sampling, histograms, clustering or representative prototypes.	Reduce rows for exploratory analysis; preserve rare failure examples through stratification.
Compression	Lossless compression for archival raw data; controlled lossy compression for selected summaries.	Reduce storage and I/O. Lossless is safer for audit and exact failure reconstruction.

6.2 Sampling methods and rare failures

Sampling method	Description	Advantages / limitations
Simple random without replacement	Select each row at most once, with equal probability.	Simple and representative for common patterns, but may select very few failures.
Simple random with replacement	A row can be selected more than once.	Useful in bootstrap estimation; duplicates may reduce unique information.
Cluster sampling	Select entire groups, e.g. machines or time blocks.	Cheaper when data is distributed by machine; may miss rare failures if selected clusters contain none.
Stratified sampling	Divide into classes/groups and sample each separately.	Ensures rare failure class is represented; proportions can be restored with weights for unbiased population estimates.

If only 0.5% of rows are failures, stratified sampling is most appropriate for model training and evaluation design. Sample failure and normal records separately, retain enough failure cases, and use class weights or training-only

resampling if needed. For final test evaluation, preserve the natural class prevalence so precision and operational performance are realistic.

6.3 Number of principal components

There is no fixed number that can be calculated from the question alone because the sensor covariance matrix is not supplied. Standardise the 200 attributes, fit PCA on training data, and retain the smallest number k that explains a chosen cumulative variance threshold (commonly 90–95%). Then validate k against failure-class recall, PR-AUC and stability. If components that explain little total variance carry rare failure signals, retain them or keep selected original attributes alongside PCA components. The objective is not variance reduction alone; it is preserving failure-detection information.

6.4 Part B – Equal-width discretization

Minimum = 3; maximum = 44; range = 41. Number of intervals = 4. Equal width = 41/4 = 10.25.

Interval	Values assigned	Count
[3, 13.25)	3, 7, 7, 9, 12	5
[13.25, 23.5)	14, 14, 15, 18, 21, 21, 22	7
[23.5, 33.75)	25, 25, 27, 31	4
[33.75, 44]	36, 36, 40, 44	4

6.5 Equal-frequency discretization

Bin	Values	Count
1	3, 7, 7, 9, 12	5
2	14, 14, 15, 18, 21	5
3	21, 22, 25, 25, 27	5
4	31, 36, 36, 40, 44	5

Equal-width bins cover intervals of the same numeric width but have unequal counts (5, 7, 4, 4). Equal-frequency bins contain the same number of observations (5 each), but their numeric widths differ.

6.6 Entropy-based supervised split

Label sequence in value order: N, N, N, N, Y, Y, N, Y, Y, Y. There are five N and five Y, so parent entropy $H(S) = -0.5\log_2 0.5 - 0.5\log_2 0.5 = 1$ bit.

Candidate split	Left class counts (N,Y)	Right class counts (N,Y)	Information gain (bits)
11.0	1, 0	4, 5	0.1080
13.5	2, 0	3, 5	0.2365
16.5	3, 0	2, 5	0.3958
20.0	4, 0	1, 5	0.6100
23.5	4, 1	1, 4	0.2781
26.5	4, 2	1, 3	0.1245
29.5	4, 3	1, 2	0.3958
33.0	4, 4	1, 1	0.2365
37.5	4, 5	1, 0	0.1080

Information Gain = H(parent) − [nL/n × H(left) + nR/n × H(right)]. The highest gain is 0.6100 bits at split point 20.0, between values 18 (N) and 22 (Y). Therefore, the best first binary split is Value ≤ 20 versus Value > 20.

6.7 Supervised versus unsupervised discretization

Equal-width and equal-frequency binning are unsupervised because they use only the distribution of the values and ignore class labels. Entropy-based discretization is supervised because it uses class labels to choose a split that reduces class uncertainty. For a classification task, entropy-based discretization is often more useful because the intervals are selected to separate classes. It must be learned using training data only to avoid test-label leakage.

Q7 – DIAGNOSE, REDESIGN AND VALIDATE (DAY-4 DESIGN CHALLENGE) (13 MARKS)

A. Diagnose the computational / pipeline defects (2 marks)

Original step	Defect and principle violated	How it misleads
1. Replace missing income with global mean	Mean imputation ignores skew, groups and missingness mechanism; preprocessing must preserve distribution and be fitted without leakage.	Pulls missing values to the centre, reduces variance and may distort risk relationships.
2. Drop occupation with 18% missing	Unjustified attribute deletion; missingness may itself contain information.	Loses potentially predictive information and may bias the population represented.
3. Equal-width income bins	Poor choice for highly right-skewed income; bins may be sparse at high incomes and crowded at low incomes.	Most observations can fall in a few bins; useful differences are hidden.
4. Min-max normalise before train/test split	Data leakage: transformation parameters use test-set information.	Test information influences training pipeline; reported generalisation can be optimistic.
5. Random 10% training sample with 3% defaults	Class imbalance and possibly inadequate training sample; random split may contain too few defaults.	Model may learn to predict non-default for almost everyone and still show high accuracy.
97% accuracy	Wrong primary metric for rare default class.	A classifier predicting every loan as non-default already achieves about 97% accuracy while detecting zero defaulters.

B. Measurable design objectives and constraints (2 marks)

- Primary objective: improve recall for the default class so a larger proportion of actual defaulters are identified.
- Also report precision, F1-score, PR-AUC, ROC-AUC, confusion matrix, specificity and calibration. Accuracy alone is insufficient.
- Set a business threshold: for example, target at least 70% default recall while keeping precision above a level agreed with the bank's review capacity. These are proposed targets and must be adjusted to actual costs.
- Avoid data leakage: split by time or use stratified split before fitting imputation, scaling, feature selection, discretization or resampling.
- Preserve realistic test prevalence; resampling must occur only inside training folds. Keep an untouched test set for final evaluation.
- Meet bank constraints: explainable decisions, reproducibility, auditability, privacy, fairness testing and acceptable scoring latency.

C. Revised preprocessing strategy (4 marks)

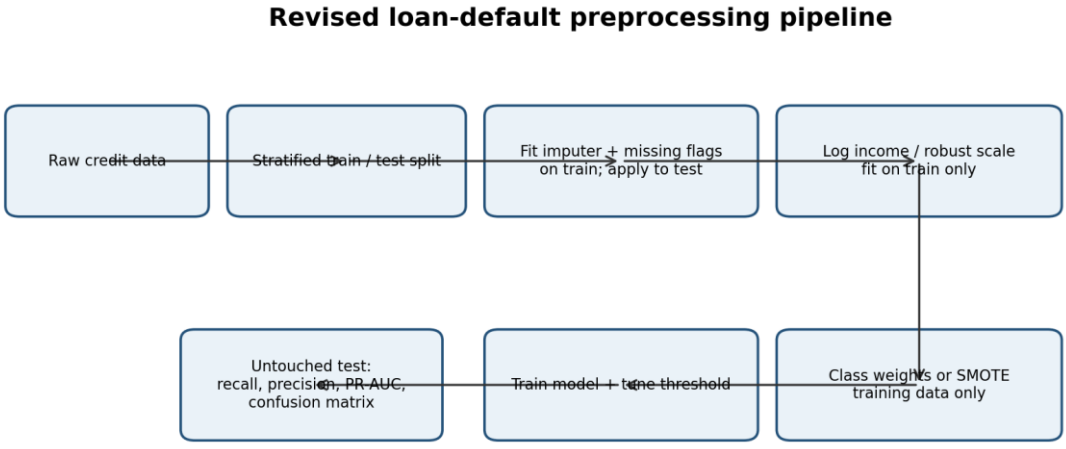


Figure 4: Recommended loan-default preprocessing and validation pipeline.

Order	Revised step	Reason and alternative considered	Fit on training only?
1	Define target and remove leakage fields; retain outcome window.	Ensure default label is defined consistently and no post-outcome variables are used.	Definition fixed before split; features checked using timestamps.
2	Split data into train, validation and test; preferably time-based and stratified where feasible.	Time split better reflects future deployment; stratification preserves rare class proportions. Never preprocess before split.	No transformation fitted yet.
3	Handle missing income with training median (or log-scale median) and add IncomeMissing flag.	Median is robust to right skew; global mean is sensitive to extreme incomes. Fit imputer on train only.	Yes; apply learned median to validation/test.
4	Retain occupation; map missing to 'Unknown' and add OccupationMissing flag. Optionally compare model-based imputation.	Dropping 18% loses information. Unknown category is transparent and safer than unsupported guessing.	Category vocabulary learned from train; apply consistently.
5	Transform skewed income using log1p; optionally use quantile transformation or robust scaling.	Log reduces right skew; equal-width bins are not suitable for heavily skewed values. Keep continuous income if tree model handles it well.	Yes; fit parameters on train only.
6	Scale numeric features using StandardScaler or RobustScaler for scale-sensitive models; tree models may not need scaling.	Robust scaling reduces outlier influence; min-max can compress most values when extremes exist.	Yes; fit scaler on train only.
7	Handle imbalance using class weights or training-only SMOTE/undersampling inside cross-validation folds.	Class weights preserve original observations; SMOTE may create synthetic patterns and must never touch test data.	Training folds only.
8	Select features using training-fold feature importance, regularisation or mutual information.	Reduces noise while preserving occupation and income signals; avoid selecting on test outcomes.	Yes; within cross-validation.
9	Train baseline and candidate models; tune decision threshold on validation data.	Compare Logistic Regression, Decision Tree / Random Forest or Gradient Boosting; threshold can trade precision against recall.	Model and threshold fitted using train/validation only.
10	Evaluate once on untouched test data and document results.	Use confusion matrix, recall, precision, PR-AUC, ROC-AUC, F1 and calibration; monitor by subgroup.	Test is used only for final evaluation.

D. Predictions before implementation (1 mark)

- Income dispersion: median imputation alone would still reduce variance, but log1p and robust treatment should reduce the influence of the high-income tail. The transformed distribution should be less right-skewed; retain the original income for interpretation.
- Training class distribution: stratified splitting preserves approximately 3% defaults and 97% non-defaults. If SMOTE is used, the training distribution becomes less imbalanced; validation and test retain the natural prevalence.
- Accuracy versus recall: accuracy may decrease slightly because the model will identify more risky loans instead of predicting the majority class for nearly all records. Recall for defaulters should improve; precision may decrease depending on threshold.

E. Implement and validate the redesign (2 marks)

Use a public credit dataset such as UCI German Credit or another approved credit-risk dataset. Ensure the target definition and class prevalence are documented. Run the original pipeline and redesigned pipeline using exactly the same held-out test records. The German Credit dataset has a different size and class balance from the scenario's 3% default rate, so report its actual prevalence and do not claim it reproduces the bank's data distribution.

Suggested Python implementation outline:

```
1. Load data and define target y (default=1, non-default=0).
```

```
2. Split train/test using a time split if dates exist; otherwise use stratified train_test_split.
3. Build a ColumnTransformer: numeric median imputation + missing indicators + log/robust scaling; categorical
most-frequent or Unknown imputation + one-hot encoding.
4. Use class_weight='balanced' where supported, or apply SMOTE only within an imbalanced-learn Pipeline on
training folds.
5. Fit candidate models and tune the classification threshold on validation data.
6. Evaluate on the untouched test set with confusion_matrix, classification_report, average_precision_score and
ROC-AUC.
```

Metric	Original pipeline (fill after run)	Redesigned pipeline (fill after run)
Accuracy		
Default-class recall		
Default-class precision		
F1-score		
PR-AUC / Average Precision		
ROC-AUC		
True positives / false negatives		
False positives		

Do not fill this comparison with guessed numbers. Include screenshots of the dataset, pipeline settings, confusion matrix and metric output from Python, Weka or KNIME. If using Python, keep the random seed and split fixed so both pipelines use the same test set.

F. Defend the final design and remaining risk (2 marks)

The redesigned pipeline is preferred because it preserves useful occupation information, treats skewed income more appropriately, prevents test-set leakage and addresses the rare default class. A decrease in accuracy is acceptable if defaulter recall and expected business value improve without overwhelming investigators with false positives. The final threshold should be selected using the relative cost of missed defaults and manual review.

One remaining risk is concept drift: borrower behaviour, economic conditions and lending policies may change after deployment. The bank should monitor default prevalence, recall from matured outcomes, calibration, feature drift and subgroup performance, and retrain only after controlled validation. Fairness and regulatory review remain necessary.

OVERALL CONCLUSION

This assignment demonstrates the use of data mining functionalities and the KDD process, along with data cleaning, integration, transformation, reduction and discretization. The numerical examples show how dispersion measures react to outliers, how missing and noisy values should be treated, how correlation and chi-square support redundancy and association analysis, and how normalisation changes distance calculations. Sampling, PCA and discretization reduce data complexity, while the loan-default challenge shows why leakage control and class-imbalance-aware evaluation are essential. All tool-based verification and Q7 model metrics should be completed with actual execution evidence.

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